

Profiling Successful Reengineering Projects

The most successful reengineering projects direct attention to social design and process transformation rather than analyzing existing procedures.

James T.C. Teng, Seung Ryul Jeong, and Varun Grover

The rapidity with which the concept of business process reengineering has been introduced and adopted by many organizations in the last few years has been nothing short of incredible. Many success stories of IT-enabled BPR have been reported [1, 7], and the enabling role of IT in BPR has been systematically examined by a number of researchers [1, 12]. With more accumulated experience, however, there is growing realization that IT is a critical enabler, but implementing reengineering involves complex sociotechnical change in an organization [2, 4]. With the “bandwagon” effect fading and reports of failures mounting, the demand for knowledge is shifting from conceptual and descriptive analysis to research results providing prescriptive guides to reengineering practice. For example, analyzing the existing business procedure has been the modus operandi for a generation of IS professionals who are accustomed to the Systems Development Life Cycle (SDLC) framework, which typically entails the computerization of existing processes (or modest changes to these processes). When we are called upon to participate in BPR projects, should we adhere to the “clean slate” approach in process redesign [7] and pay less attention to the analy-

sis of existing business procedures? If fundamental redesign is called for, do we implement the new design through radical changes to the organization? Or perhaps it is better to attempt incremental adjustments that may be easier to manage [11].

In this article we seek to provide substantive answers to these questions based on empirical evidence collected from more than 100 BPR project participants. This is accomplished by examining the relationships between characteristics of the reengineering project (referred to as “project profile”) and implementation success. Specifically, two aspects of the project profile will be examined: radicalness of the project and the strength of effort spent on the various stages of the project (referred to as “stage-efforts”). As shown in Figure 1, the proposed research model relates the two independent variables regarding project profile to reengineering implementation success in order to answer these important questions:

- Are reengineering projects aimed at more radical change resulting in higher implementation success?
- If limited attention and resources must be allocated among the different stages of a reengineering project, which stage (or stages) should receive more emphasis in order to achieve higher implementation success?

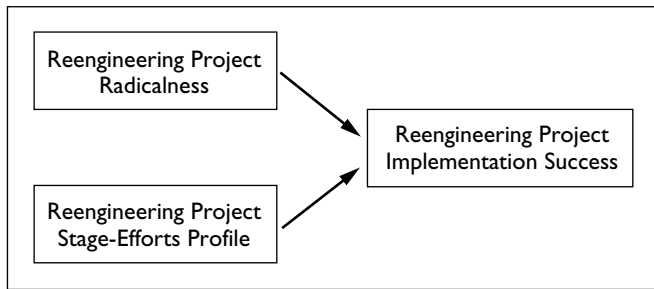


Figure 1. Research model

Measurement of Research Variables

To measure the degree of radicalness of a reengineering project, we adopt the multidimensional view commonly ascribed to by BPR practitioners and researchers. Hammer and Champy [7], for instance, presented this idea in the form of a “reengineering wheel” that includes changes in jobs, structures, management and measurement systems, and values and beliefs. The set of field case studies reported by Hall, Rosenthal, and Wade [5] has demonstrated that, to achieve performance breakthrough, the redesign should fundamentally change a spectrum of related organizational elements: 1) roles and responsibilities, 2) measurements and incentives, 3) organizational structure, 4) information technology, 5) shared values, and 6) skills. Based on this work, we measure the degree of BPR project radicalness along seven dimensions—six of these were drawn from

Hall, et al. [5] as listed previously, and the seventh dimension pertains to change to the patterns of process work flow. Respondents were asked to “assess *the extent of change* your team has attempted in redesigning the process in terms of the following seven dimensions (1=moderate change, 3=substantial change, 5=radical change).”

Although not all responding companies are expected to use formal methodologies, both formal and informal approaches to BPR projects have several generic stages, and certain tasks are typically performed within each stage [1, 8]. To evaluate the strength of efforts spent in various stages of the project (stage-efforts), an eight-stage framework was adopted based on a number of published reengineering methodologies (e.g., [9]). The respondents were presented with the outline of the eight-stage framework with a set of specific tasks within each stage, as shown in Table 1. The instruction was “for each stage, please enter a number between 1 and 5 to indicate the *strength* of the effort that your project team has attempted to complete those tasks (1=very weak, 2=weak, 3=medium, 4=strong, 5=very strong).” Further, they were instructed to leave the box blank “if the stage was not attempted.”

Reengineering project success was measured multidimensionally based on two different perspectives: perceived level of success [3] and goal fulfillment [6, 10]. In this study, respondents were asked to answer

Table 1. Reengineering project stages and tasks

Stage 1. Identification of BR Opportunities Establish a steering committee for overall BR planning Secure management commitment Align with corporate and IT strategies Identify major business processes with a “business model” Understand customers’ requirements Prioritize processes and select one for implementation	Stage 5a. Solution: Technical Design Develop and evaluate alternative process designs Detailed process modeling (entities, relationships, etc.) Design controls for process integrity IS analysis and design for the new process Prototype and refine the process design
Stage 2. Project Preparation Plan for organizational change (e.g., inform stakeholders) Organize a BR team for the selected process Train the BR team members Conduct project planning	Stage 5b. Solution: Social Design Empower customer contact personnel Define jobs and incentives Develop and foster shared values Define skill requirements and career paths Design new organizational structure Design employee performance measurement schemes Design change management program
Stage 3. Analysis of the Existing Process Analyze existing process structure and flows Identify value-adding activities Identify opportunities for process improvement	Stage 6. Process Transformation Develop test and rollout plans Implement the social and technical design Train staff and pilot new process
Stage 4. Development of a Process Vision Understand process customers’ requirements Identify process performance measures Set process performance goal Identify IT that enables process redesign Develop a vision (preliminary sketch) for the redesigned process	Stage 7. Process Evaluation Monitor performance Continuous improvement

Table 2. Respondents by industry**(a) Overall Respondents**

Industry Type	Frequency	Percent
Manufacturing	74	31.0
Banking/Finance	21	8.8
Insurance	20	8.4
Distribution	16	6.7
Utility	14	5.9
Service	12	5.0
Information Services	12	5.0
Healthcare	12	5.0
Marketing	7	2.9
Construction/Engineering	7	2.9
Telecommunications	6	2.5
Transportation	5	2.1
Government	5	2.1
Retail	4	1.7
Food Processing	3	1.3
Others	15	6.3
No answer	5	2.1
Total	239	100.0

(b) Reengineering Firms

Industry Type	Frequency	Percent
Manufacturing	30	28.6
Banking/Finance	14	13.3
Insurance	11	10.5
Utility	7	6.7
Telecommunications	6	2.5
Information Services	5	4.8
Distribution	4	3.8
Service	4	3.8
Construction/Engineering	3	2.9
Government	3	2.9
Retail	3	2.9
Marketing	2	1.9
Food Processing	2	1.9
Others	10	9.5
No answer	1	1.0
Total	105	100.0

one five-point scaled question about the perceived success level of their project. The goal fulfillment perspective can be characterized as objective compared to the subjective nature of perceived success. This perspective has frequently been the basis of studies in evaluation research [10] and the MIS literature [6]. A question based on this perspective would be: To what extent is the goal(s) of reengineering fulfilled? This comparison is particularly meaningful as a success measure for BPR since reengineering has been conceptualized as a deliberate change initiative aimed at 'breakthrough' performance gains, and the setting of performance improvement goals is central to the BPR concept. The present study used five commonly emphasized BPR goals: 1) cost reduction, 2) cycle-

time reduction, 3) customer satisfaction level increase, 4) worker productivity increase, and 5) defects reduction [1]. For each goal, respondents indicated the planned level of performance improvement (before reengineering) and the actual level of improvement achieved (after project implementation). They were also directed to leave the spaces blank if the presented goal was not applicable to their projects. For each performance indicator, we compute a "ratio of fulfillment," which is the actual level of improvement divided by the planned level.

Research Instrument and Sample

To be sure the respondents correctly comprehend the meaning of BPR and did not mistake TQM or MIS projects as reengineering initiatives, BPR was carefully defined at the beginning of the questionnaire as a deliberate (planned) change, typically enabled by IT in an attempt to redesign a business process to achieve performance breakthroughs in quality, speed, customer service, and cost. It was also pointed out that reengineering has often been contrasted with automation, which mainly involves computerization of existing business processes. Respondents were asked to select a recently concluded reengineering project in which they had participated as a team member as the basis for answering all items on the questionnaire. They were also requested to describe the project, the process being reengineered, and the performance criteria used. This detailed characterization of their efforts provides a high level of assurance that the projects attempted are indeed reengineering initiatives.

Because of the top-down strategic nature of most reengineering endeavors, professional business planners can be expected to have a high level of interest and involvement in BPR. With this in mind, the researchers requested and were granted permission to contact members of Planning Forum, a professional association focusing on strategic management and planning. A total of 853 questionnaires were sent, and a part of the questionnaire may be completed even if no reengineering project had been conducted. A total of 239 usable responses were returned, resulting in a response rate of 28%. To determine whether the respondents reflected the sample frame of Planning Forum members, non-response bias was assessed. Early respondents were compared to late respondents across a number of key organizational characteristics—distribution of industry type, number of employees, company annual sales, organization type, etc. None of the Chi-squares or T-tests showed significance, providing support for generalizability of the sample frame.

Table 3. Correlation analysis for radicalness and success

Project Success	Patterns of process work flows	Roles and responsibilities	Performance measurements and incentives	Organizational structure	Information technology applications	Shared value (culture)	Skill requirements	Overall extent of change
Overall success level	.427 ***	.324 ***	.351 ***	.102	.280 ***	.173*	.166	.409 ***
Cost reduction[#]	.269 * .092	.159 .194	.231 .093	.260 * .030	.165 -.000	.139 .058	.129 .022	.291 ** .110
Cycle time reduction[#]	.191 .127	.134 .050	.198 .194	.033 -.157	.033 -.021	.111 .029	.098 .008	.171 .052
Customer satisfaction increase[#]	.258 .242	.187 .129	.282 * .366 ***	.180 -.242	-.022 .039	-.019 .024	-.112 -.168	.182 .098
Worker productivity increase[#]	.122 .061	.107 .197	.159 .431 ***	.131 .069	.011 -.187	.047 .053	-.043 -.132	.125 .134
Defects reduction[#]	-.124 -.073	-.041 .109	.151 .044	.113 .011	-.091 -.380 *	.058 -.133	-.218 -.272	-.015 -.173

* Significance at $p < .10$ ** Significance at $p < .05$ *** Significance at $p < .01$

The first figure is the correlation with actual level of improvement, the second figure, the ratio of fulfillment.

Responding Firms and Their Projects

Of the 239 respondents, 105 (44%) had completed at least one BPR project (referred to as “reengineering firms”) and were able to respond to the entire questionnaire. As shown in Table 2, almost two-thirds of the responding firms were either in the manufacturing industry, the financial sector, or the service industry. The remaining one-third are involved in more than 10 different industries. The distribution of the reengineering firms was similar to that of total responses, with a majority of them in manufacturing and banking/finance/insurance (52.4%). Organization size was measured by the number of employees: The data suggests that larger companies are more likely to implement reengineering projects. For the 105 reengineering firms, 25 have over 10,000 employees (23.8%), 17 have between 5,000 to 10,000 employees (16.2%), and 29 have 1,000 to 5,000 employees (27.6%).

Among the 105 processes reengineered in the selected projects, the three most popular business processes were: customer service (13), product development (13) and order management (10). Two of these processes—customer service and order management—involve customer-interfacing activities. This is not surprising, as it has been widely recognized that processes at the customer interface are often essential to a firm’s cash flow and customer satisfaction. Next on the list were: business planning and analysis (5.7%), financial systems (4.8%) and accounting processes (3.8%). Several other processes were also reengineered (e.g., service delivery (2.9%), distribu-

tion (2.9%), purchasing (2.9%), etc.). It was also found that firms put emphasis on several industry-specific processes (e.g., product development and order management processes for manufacturing firms; financial systems and loan management processes for banking/finance firms; and claims processing and new business underwriting processes for insurance firms), indicating that more organizations are now focusing on processes critical to their business performance. However, among the three most popular target processes, customer service processes appear to be common across most industries.

While a majority of the companies (67.6%) used a formal methodology to guide their efforts, others (29.5%) did not. Of the 71 companies that used a methodology, only 16 (22.5%) had developed the methodologies in-house. In 26 cases (36.6%), the methodology was provided by a consulting group and 29 firms (40.8%) used a consultant’s methodology modified for their company.

We asked the reengineering firms to identify the information technology that played an important enabling role in their projects. Most respondents indicated that they utilized databases. Databases are inherently useful for cross-functional reengineering since different departments can share a common information repository to eliminate redundant tasks and coordinate their activities [12]. Telecommunication networks, used by 38.1% of the respondents, further affirm the critical needs in BPR to break down walls between traditional functions and to overcome geography [1]. Many companies also used work-flow soft-

Table 4. Correlation analysis for project stage-efforts and implementation success

Project Success	Identification of BPR opportunities	Project preparation	Analysis of the existing process	Development of a process vision	Solution: technical design	Solution: social design	Process transformation	Process evaluation
Overall success level	.247 **	.244 **	.139	.242 **	.199 *	.390 ***	.432 ***	.547 ***
Cost reduction[#]	.134 -.116	.103 .205	.013 .269 *	.095 .188	.165 -.000	.314 ** .203	.220 .339 **	.386 *** .577 ***
Cycle time reduction[#]	.039 -.126	.042 .122	.072 .196	.219 .225	.033 -.021	.274 ** .108	.267 * .222	.342 ** .455 ***
Customer satisfaction increase[#]	.248 .122	-.064 .147	.156 .280 *	.232 .392 **	-.022 .039	.354 ** .360 **	.211 .310 **	.374 ** .404 **
Worker productivity increase[#]	.092 .000	.080 .115	.038 .261 *	.214 .193	.011 -.187	.294 ** .470 ***	.213 .299 *	.314 ** .619 ***
Defects reduction[#]	.184 -.025	-.171 .154	.020 .285	.172 -.064	-.091 -.380 *	.399 ** .071	.257 .275	.351 * .551 ***

* Significance at $p < .10$ ** Significance at $p < .05$ *** Significance at $p < .01$

The first figure is the correlation with actual level of improvement, the second figure, the ratio of fulfillment.

ware (21%) and imaging (9.5%) to enable BPR, while a few mentioned EDI, client-server and expert systems in their efforts. It is revealing to note that the two most frequently mentioned enablers, databases and telecommunication networks, can be predicted on the basis of a theoretical framework proposed by Teng, Grover, and Fiedler [12]. In general, the set of enablers fulfill the various needs of BPR identified by Davenport and Short [1].

The average strength of effort (on a scale of 1 to 5) showed the following stage-effort pattern: Stage 3 (3.94), Stage 1 (3.80), Stage 4 (3.63), Stage 2 (3.46), Stage 6 (3.39), Stage 5a (3.37), Stage 7 (3.21), and Stage 5b (3.09). The results show that the stage that received the most emphasis is the analysis of the existing process stage, indicating that the “clean slate” approach to BPR is rarely found in practice. This is not unexpected given the tangible nature of the existing process and the need to identify redundant tasks in the current procedure. It appears that most reengineering practitioners believe it is prudent to thoroughly comprehend the existing process before developing visions for a new one. The next most emphasized stage was “identification of reengineering opportunities,” followed by the stages of “vision,” “project preparation,” “process transformation,” “technical design,” and “process evaluation.” Surprisingly, the sample firms focused least on the social design stage, while the literature emphasized the importance of designing change management programs, incentives, measurement systems, developing

values, and defining skill requirements [5, 11]. It should also be noted that the first four stages received the four highest stage-effort scores, indicating that firms usually put more emphasis on early rather than later stages of reengineering projects.

Project Radicalness and Implementation Success

The results of correlation analysis between radicalness and reengineering success are presented in Table 3. For perceived level of success, we found strong positive relationships between radicalness and success. Of the seven change dimensions, four—work flow patterns, roles and responsibilities, performance measurements and incentives, and IT applications—were significant at the 0.01 level and one—shared values (culture)—was significant at the 0.10 level. Although both skill requirements and structure dimensions failed to have significant associations, overall radicalness (the average of the seven dimensions) was highly correlated with success ($r=0.409$, $p<0.01$). Thus, the findings generally support the contention that *likelihood of reengineering success increases with more thorough and substantial change to the key elements of the organization* [5].

For the goal-fulfillment indicators, we found only a few significant associations. However, it is revealing to note and interpret how certain improvement goals are associated with specific changes. First, some support was found for the relationship between cost reduction and changes in work patterns and organiza-

tional structure. Current patterns of work flow and organizational structure, often in the form of redundant processing steps and personnel, may have caused many cost-related problems, and become targets for radical changes, especially when cost reduction is a primary goal. While these relationships were found

eight) were significantly associated with perceived overall success. While we found moderate effects in identification, preparation, and vision stages, the strongest effects were observed in the social design, process transformation, and process evaluation stages. This confirms the importance of change management and human resource management programs

Table 5. Stage-efforts vs. impact on perceived project success

Stage	Average effort [#]	Correlation with perceived success [#]
Stage 3: Analysis of the Existing Process	3.94 (1)	.139 (8)
Stage 1: Identification of BPR Opportunities	3.80 (2)	.247** (4)
Stage 4: Development of a Process Vision	3.62 (3)	.242** (6)
Stage 2: Project Preparation	3.46 (4)	.244** (5)
Stage 6: Process Transformation	3.39 (5)	.432*** (2)
Stage 5a: Solution: Technical Design	3.37 (6)	.199* (7)
Stage 7: Process Evaluation	3.21 (7)	.547*** (1)
Stage 5b: Solution: Social Design	3.09 (8)	.390*** (3)

[#] Numbers in parentheses indicate ranking

for actual cost reductions, they did not hold for ratio of fulfillment, possibly due to overly ambitious goals. Second, radical change in performance measurements and incentive systems were related to worker productivity and the level of customer satisfaction. These findings suggest that by fundamentally altering measurement and incentive systems, companies can more accurately measure employees' performance and reward them on the basis of the value they add in serving customers, thus increasing productivity and customer satisfaction.

A somewhat surprising finding is the lack of relationship between cycle time reduction and any change dimensions. The data suggests that cycle-time improvement tends to reach a certain plateau and stay there regardless of the magnitude of the change effort. One possible interpretation is the physical limitation of how fast a process can be completed. Even with drastic changes to a process, there is always a minimum of time needed to accomplish the tasks required. There was also a lack of relationship between defect reduction and the various change dimensions. In fact, the possibility exists that a negative relationship may develop, as in the case between changes in IT application and the ratio of fulfillment in defect reduction. It seems likely that modest change is optimal for defect reduction, and too much change may lead to chaos and more possibility for mistakes. Future research should further investigate these curious findings.

Project Stage-Efforts Profile and Implementation Success

As can be seen in Table 4, most stages (seven out of

in BPR. The findings also highlight the significance of later stages of reengineering, particularly continuous monitoring and evaluation, for BPR success.

Interestingly, we found that the effort directed at analyzing existing processes did not affect the overall success of implementation. BPR seeks fundamentally to *change* existing processes to improve efficiency and effectiveness, and the lack of impact of

effort spent on analyzing the existing processes is certainly logically consistent with that very concept. However, this is the first piece of empirical evidence demonstrating this important point. The results, however, do not suggest this project stage should be eliminated. The findings do indicate that overemphasis on analysis of existing business processes is unlikely to make BPR implementation significantly more successful.

The potential impact of technical design effort on success, although significant, is the weakest. This lends further support to recent realization among IS practitioners and researchers that process reengineering is, first and foremost, structural and organizational change [2, 4]. Information technologies are important and often essential enablers, but the implementation involves multidimensional organizational change. Technical design effort appears to offer very moderate relief to these problems and make a limited contribution to project success.

Several significant associations with goal-fulfillment measures are shown in Table 4. The most impressive finding is the strong effects of the last three stages (social design, process transformation, and evaluation) on the various goal-fulfillment indicators. Other stages had only a few associations, and they were weaker than those for the last three stages. These results imply that programs for change management, human resource management, and continuous improvement are critical for success with respect to any goal-fulfillment criteria. These findings reinforce recent empirical study results reported by Grover et al. [4], indicating that change management was regarded as the most severe reengineering imple-

mentation problem by project participants. In addition, they also found that, while both change management and technological competence problems were regarded as very difficult by project participants, the former was related much more strongly to reengineering success than the latter [4]. Recent thinking in the field is evolving from a “one-shot” reengineering to the notion of on-going “process management.” [2]. Our research findings provide empirical support for the validity of this concept.

Discussion

Findings from the study have significant implications for both research and practice. The strong positive relationships between radicalness and implementation success suggest that likelihood of reengineering success increases with more thorough and substantial change of the key elements of the organization. It is important, however, not to equate “radical” process change, as described here, with the reckless overhaul of a process without considering its human and organizational ramifications. In fact, the study shows that in addition to changes in work flow patterns and the application of IT, organizational elements such as the roles and responsibility as well as performance measurement and incentives are also highly related to perceived success. IS and BPR professionals can increase the prospects of project success through substantial changes to both the technical and the social components of the process.

Further insights into the importance of social and organizational elements in BPR can be gained by examining both the average stage-efforts and the impact of these efforts on perceived project success (Table 5). By juxtaposing these two results, as can be seen in Table 5, an interesting pattern emerges: Extent of effort devoted to a stage appears to be inversely related to its potential impact on perceived project success. For example, stage 3 (analysis of the existing process) received the most effort on average, but it has no significant correlation with perceived success. On the other hand, stages 7 and 5b (process evaluation and social design) received the least amount of effort, but ranked on top in terms of potential impact on both perceived success and specific goals. These results strongly suggest:

- IS and BPR professionals have not paid sufficient attention to the later stages of reengineering projects such as social design, process transformation, and evaluation;
- Additional effort directed at these later stages will likely be more effective than efforts at other stages in improving the odds of project success.

As the number of reengineering failures increases, these results provide specific reengineering project implementation guidelines for IS and BPR professionals based on solid empirical research rather than consultants’ subjective opinions.

The study has generated a number of interesting and informative findings. A large number of reengineering projects were included in the sample and these results should have more generalizability than small-sample case studies [5]. The results, however, should be interpreted with caution. The correlations between project profile and success, for instance, do not imply causality. The projects selected by the respondents for assessment also may not be representative of the variety of BPR initiatives attempted. In this study we undertook a preliminary step in attempting to understand a multifaceted complex phenomenon involving dynamic and fundamental change. With the management of change becoming a key element of organizational life, it is important to delve further into these issues adopting a variety of methods of inquiry.

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JAMES T.C. TENG (fsujteng@darla.badm.sc.edu) is an associate professor at the College of Business Administration, University of South Carolina.

SEUNG RYUL JEONG (srjeong@kookmin.ac.kr) is on the faculty of the College of Economics and Business Administration at Kook Min University, Korea.

VARUN GROVER (vgrover@darla.badm.sc.edu) is an associate professor at the College of Business Administration, University of South Carolina.

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